

Sampling practice handbook

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Stratified sampling

In stratified sampling, a population is divided into nonoverlapping strata, and a sample is selected within each stratum. With proportional allocation, the sample share for a stratum equals its population share. For two strata of 800 and 400 units and a total sample of 120, proportional allocation selects 80 and 40 units respectively.

Stratified sampling

For a stratified population mean, stratum sample means are combined using the population shares of the strata. If two strata contain 800 and 400 units, their population shares are two-thirds and one-third. A simple unweighted average of the two stratum means is generally not the same as this population-weighted mean.

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Sampling weights

A basic design weight is the reciprocal of a unit's selection probability. If a unit has selection probability 0.10, its design weight is 10. A design weight represents how many population units the sampled unit represents under the sampling design; it is not a score for the quality of a response.

Frames and coverage

A sampling frame should identify eligible sampling units without duplicates or omissions. Duplicated units can receive unintended selection opportunities. Missing eligible units create undercoverage. These are frame-quality problems and cannot be fixed merely by increasing the sample size from the same defective frame.

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Strata and clusters

Stratification selects samples within every stratum. In one-stage cluster sampling, a sample of clusters is selected and all units in those selected clusters are observed. These designs differ in which groups are selected and which units within those groups are observed.